PROJECT REPORT

# Multimodal Autism Screening System

Phase-wise development summary: from emotion recognition and behaviour detection to multimodal risk prediction with XGBoost

## Project at a glance

| | | | |
|---|---|---|---|
| 1 | **Emotion Recognition** | FER2013 · EfficientNet-B0 | `emotion_model.pth` |
| 2 | **Autism Face Analysis** | Autism Face Dataset · ResNet50 | `autism_face_model.pth` |
| 3 | **Repetitive Behaviour Detection** | SSBD · MediaPipe Pose + LSTM | `behavior_model.pth` |
| 4 | **Audio Analysis** | Whisper · Silero-VAD · OpenSMILE | `audio_features.csv` |
| 5 | **Questionnaire Scoring** | M-CHAT-R (20 questions) | `Q1–Q20 indicators` |
| 6 | **Multimodal Fusion** | XGBoost | `4 risk predictions` |

# 1. Project Overview

This report documents the phase-wise development of a multimodal screening pipeline for early behavioural risk assessment in children. Six phases extract complementary signals: facial emotion, facial attention, repetitive movements, speech and audio biomarkers, and a parent-reported questionnaire. These signals are fused into a single feature row per child and passed to an XGBoost model that predicts four risk categories.

## System architecture

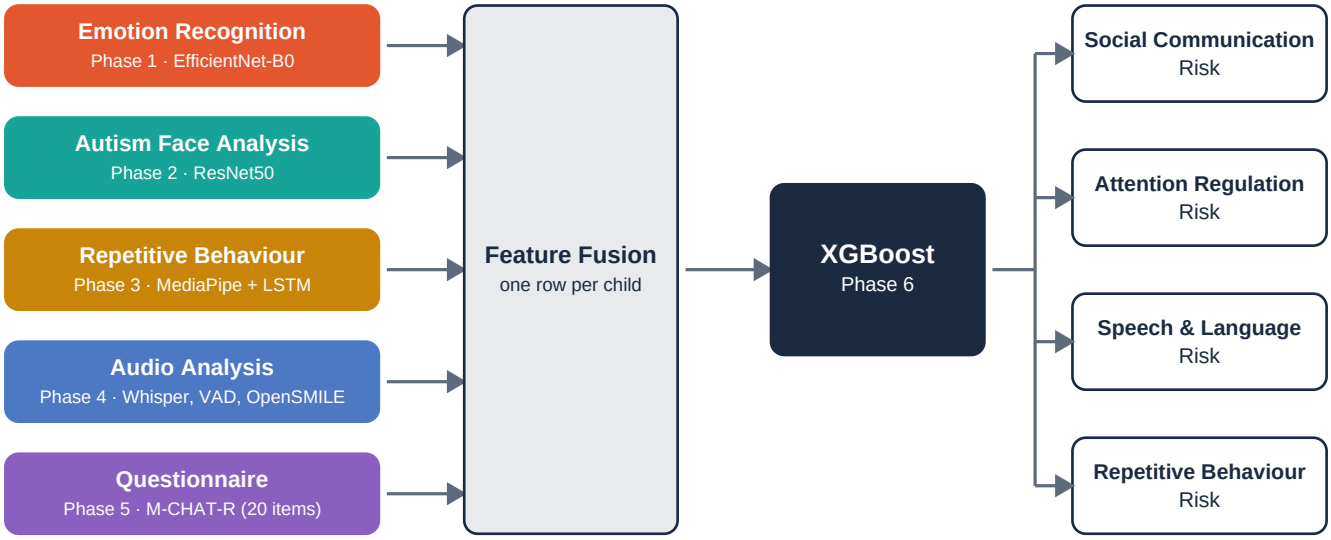



Figure 1. Data flow from the five feature-extraction modules through fusion to the final XGBoost predictions.

## Phase-wise division

| Phase | Module | Dataset / Tools | Model | Output |
|---|---|---|---|---|
| 1 | Emotion Recognition | FER2013 | EfficientNet-B0 | emotion_model.pth |
| 2 | Autism Face Analysis | Autism Face Dataset | ResNet50 | autism_face_model.pth |
| 3 | Repetitive Behaviour Detection | SSBD | MediaPipe Pose + LSTM | behavior_model.pth |
| 4 | Audio Analysis | Whisper, Silero-VAD, OpenSMILE | Pre-trained (no training) | audio_features.csv |
| 5 | Questionnaire Scoring | M-CHAT-R | Rule-based scoring | Q1–Q20 indicators |
| 6 | Multimodal Fusion | Fused feature table | XGBoost | 4 risk predictions |

PHASE 1

# Emotion Recognition

Train FER2013 emotion model

| | |
|---|---|
| **Dataset** | FER2013 |
| **Model** | EfficientNet-B0 (ImageNet pre-trained, fine-tuned) |
| **Output** | `emotion_model.pth` |
| **Detects** | 7 emotion classes |

## Approach

Rather than training from scratch, the model starts from ImageNet pre-trained weights and is fine-tuned on FER2013. Transfer learning with EfficientNet-B0 is the standard approach for custom image classification tasks and is far more practical on Google Colab. The detailed execution steps are documented in `process.txt`.

- Pre-trained ImageNet features give a strong starting point for facial-expression classification.
- Fine-tuning adapts the network to the seven FER2013 emotion classes.
- The trained weights are saved as `emotion_model.pth` for downstream use.

## Emotion classes

Happy | Sad | Angry | Fear | Neutral | Surprise | Disgust

## Results

**64.35%**
Test Accuracy

**64.27%**
Test Weighted F1

PHASE 2

## Autism Face Analysis

Train autism face model

| | |
|---|---|
| **Dataset** | Autism Face Dataset |
| **Model** | ResNet50 |
| **Output** | `autism_face_model.pth` |
| **Process** | Evaluates facial features to identify risk probabilities |

### Features extracted

- Eye Contact
- Face Attention
- Face Focus

### Results

**88.00%**
Best Validation Accuracy

**85.50%**
Final Test Accuracy

**0.8539**
Test Macro F1

PHASE 3

## Repetitive Behaviour Detection

Train SSBD model

| | |
|---|---|
| **Dataset** | SSBD (Autism Behaviour Video Dataset) |
| **Classes** | armFlapping, headBanging, spinning |
| **Output** | `behavior_model.pth` |

### Pipeline

Figure 2. Behaviour-classification pipeline. Body pose is tracked per frame and the landmark sequence is classified by an LSTM.

### Behaviour classes

- armFlapping
- headBanging
- spinning

### Results

**~91.89%**
Validation Accuracy

**0.9195**
Macro F1

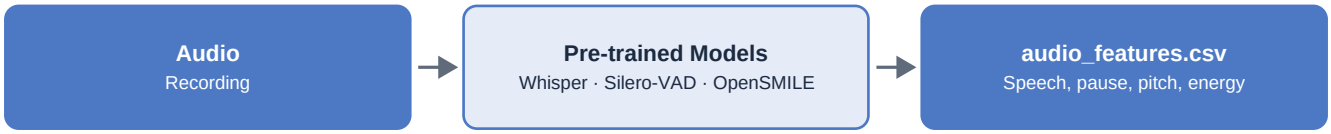
PHASE 4

## Audio Analysis

Audio biomarker extraction

| | |
|---|---|
| **Process** | No training required. Pre-trained models are used to extract audio biomarkers. |
| **Tools** | Whisper, Silero-VAD, OpenSMILE |
| **Output** | `audio_features.csv` |

### Pipeline




### Features extracted

| | | | | | |
|---|---|---|---|---|---|
| Speech Rate | Pause Count | Pause Duration | Response Latency | Pitch | Energy |

PHASE 5

## Questionnaire Scoring

M-CHAT-R questionnaire module

| | |
|---|---|
| **Instrument** | M-CHAT-R (Modified Checklist for Autism in Toddlers, Revised) |
| **Questions** | 20 screening questions |
| **Scoring** | Mapped to established clinical scoring rules |
| **Fusion input** | Q1–Q20 as binary indicators (0 / 1) |

Each of the 20 answers is encoded as a binary indicator so it can be fused with the other modalities in Phase 6. For reference, the standard M-CHAT-R total-score bands are: **0–2** low risk, **3–7** medium risk, **8–20** high risk.

PHASE 6

# Multimodal Fusion (XGBoost)

Feature fusion pipeline and final model

The fusion module aggregates the features extracted across all modalities into **one final row per child**, enabling comprehensive behavioural reasoning across every signal.

## Example feature array

| Feature | Source | Example value |
|---|---|---|
| Emotion Score | Phase 1 · Emotion | 0.75 |
| Eye Contact | Phase 2 · Face | 0.42 |
| Arm Flapping | Phase 3 · Behaviour | 14 |
| Head Banging | Phase 3 · Behaviour | 2 |
| Spinning | Phase 3 · Behaviour | 6 |
| Speech Rate | Phase 4 · Audio | 82 |
| Avg Pause | Phase 4 · Audio | 1.6 |
| Response Latency | Phase 4 · Audio | 3.2 |
| Q1–Q20 | Phase 5 · Questionnaire | Binary indicators (e.g. 0, 1) |

## Final XGBoost model

| | |
|---|---|
| **Input** | All fused features (Phases 1–5) |
| **Output** | Four risk predictions, listed below |

**Social Communication Risk**
Social interaction and communication cues

**Attention Regulation Risk**
Attention and focus patterns

**Speech & Language Risk**
Speech and language patterns

**Repetitive Behaviour Risk**
Repetitive movement patterns

# Results Summary

Reported performance of the trained models. Phases 1 and 2 are reported on the test set; Phase 3 figures are validation-set results.

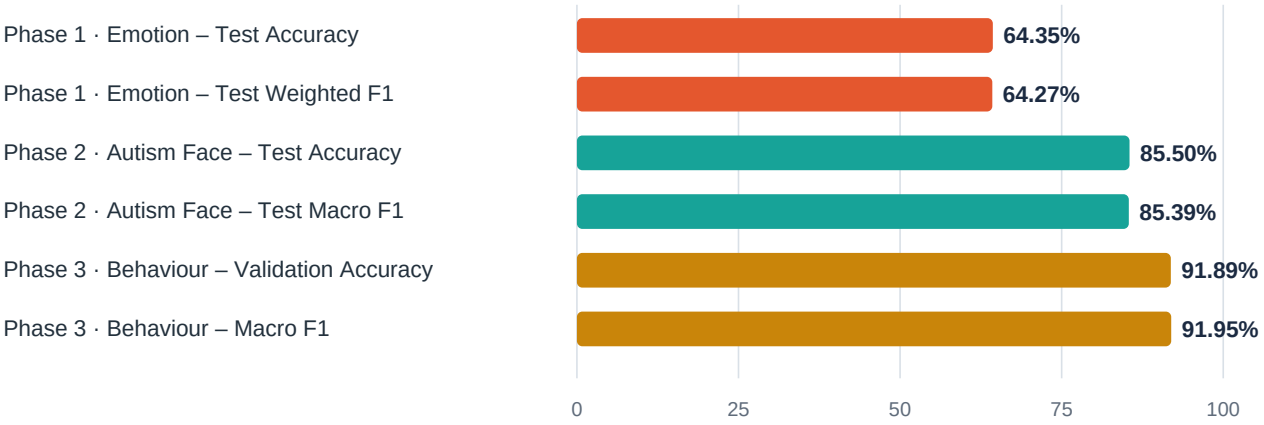



Figure 3. Reported metrics by phase (%). F1 values of 0.8539 and 0.9195 are shown as percentages.

## Deliverables

| Artifact | Phase | Description |
|---|---|---|
| `emotion_model.pth` | 1 | Fine-tuned EfficientNet-B0 weights for 7-class emotion recognition |
| `autism_face_model.pth` | 2 | ResNet50 weights for facial risk analysis |
| `behavior_model.pth` | 3 | LSTM weights for repetitive behaviour classification |
| `audio_features.csv` | 4 | Speech and acoustic biomarkers per child |
| Q1–Q20 indicators | 5 | Binary M-CHAT-R item responses |
| XGBoost fusion model | 6 | Predicts four risk categories from the fused feature row |

## Notes and next steps

- **Fusion evaluation:** performance metrics for the Phase 6 XGBoost model are not yet included in this report and should be added once evaluation is complete.
- **Emotion model:** FER2013 is a challenging benchmark; the 64.35% test accuracy leaves room for improvement (e.g. augmentation or class balancing).
- **Behaviour and face models:** the Phase 3 results are validation-set figures; a held-out test evaluation would make them directly comparable with Phases 1 and 2.
- **Responsible use:** this system is intended as a screening and decision-support aid. Its outputs are risk indicators and do not constitute a clinical diagnosis; results should be reviewed by qualified professionals.